INCH-POUND
MIL-PRF-53137A(CR4)
AMENDMENT 1
21 April 2000

PERFORMANCE SPECIFICATION

HEATER, ARMY SPACE (ASH), ELECTRIC POWERED, MULTIFUEL

This amendment is approved for use within the USA Communications-Electronics Command, Logistics and Readiness Center, Department of the Army, with MIL-PRF-53137A(CR4)), dated 10 July 1997.

PAGE 2

2.2.1, under STANDARDS, DEPARTMENT OF DEFENSE, delete "MIL-STD-462" in its entirety.

Add:
“MIL-STD-2169B - High-Altitude Electromagnetic Pulse (HEMP) Environment.

(Copies of the above classified military standard are available to personnel with the appropriate security clearance from Headquarters US Army Communications-Electronics Command, Research, Development and Engineering Center, Command/Control and Systems, Integration Directorate - South, ATTN: AMSEL-RD-C2-PD-E, 10108 Gridley Road, Suite 1, Ft Belvoir, VA 22060-5817.)”

2.2.2, last paragraph, at the end of the paragraph: change "AMSEL-LC-IEW-D-EE" to "AMSEL-LC-CCS-G-GB".

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3.1, first sentence: delete "with either diesel or JP-8 fuels" and substitute "with both diesel and JP-8 fuels".

At the end of the paragraph, add the following sentence: "Thresholds and objectives are identified in this document by [T] and [O], respectively."

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3.3.1, delete the first sentence and substitute: "The heater shall deliver not less than 120,000 [T], 150,000 [O] British Thermal Units per Hour (BTUH) of clean air (see 3.8.7) when operated in the all fresh air mode, with an air temperature of 0 °F, at sea level up to 4,000 feet altitude, against an 0.5-inch water gauge (WG) pressure, using diesel fuel, grade Low Sulfur No. 1-D conforming to A-A-52557."

3.3.2: Add the symbol "[T]" after each dimension in this paragraph.

Add the following sentence at the end of the paragraph: "Producing heaters that are smaller than the above dimensions is an objective of this document."

3.3.3: line 1, delete "The heater shall weigh no more than 355 pounds " and substitute: "The heater shall weigh no more than 360 pounds [T], 355 pounds [O], including ".

3.4.1, line 3, delete "from -60 to +160 °F ", and substitute: "from -60 [T], -65 [O] to +160 °F".

3.4.2: line 3, delete "from -40 to +65 °F" and substitute: " from -40 [T], -60 [O] °F unassisted start, -60 [T], -65 [O] °F assisted start to +65 °F, in all climatic conditions."

3.4.6: lines 4 and 5, delete "shall not exceed 85 db(A)" and substitute: "shall not exceed 75 db(A) [T], 65 db(A) [O] ".

Lines 5 and 6, delete "shall not exceed 65 db(A)." and substitute: "shall not exceed 65 db(A) [T], 55 db(A) [O]."

3.4.7, delete in its entirety and substitute:

"3.4.7 NBC compatibility and High-Altitude Electromagnetic Pulse (HEMP). The heaters will be mission essential in many applications; therefore, NBC contamination survivability is required. The heaters will be decontaminable to negligible risk levels, will be hardened against the effects of chemical agents and decontaminants, and will be compatible with soldiers in full NBC-protective ensemble. The heaters must be HEMP survivable but operation through a HEMP event is not required. The recycling of power to restore operations after a HEMP event is acceptable. When tested as specified in 4.6.8.2, the heater shall operate normally following a HEMP event (see 6.9)."

3.5.1, first sentence: Delete "figure 1" and substitute "figure 1-a."

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3.5.1, at the end of the paragraph add new sentence: "Conducted emissions (CE102) on the heater power leads shall not exceed the applicable values shown on Figure 1-b."

3.5.3, at the end of the last sentence: delete "cover." and substitute "cover (see 6.8)."

3.6, at the beginning of the paragraph add: "The heaters shall primarily operate using JP8 and diesel fuels DF-1, DF-2, and DF-A, as alternative fuels. The fuel system shall be easily accessible and allow for rapid draining."

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3.7.1, line 9, seventh sentence: delete: "and self concealing."

3.7.5, delete entire paragraph and substitute:

"3.7.5 Mobility/parking devices. The heaters are required to be soldier-portable for short distances for repositioning purposes similar to the heaters currently in the field. The heater shall be mounted on a chassis with integral skids and tires, and the tires shall be mounted on wheels that are removable or retractable without the use of external tools. It is desirable to have wide, ski-like skids on the ASH for movement over mud and snow. It is also desirable to utilize non-pneumatic (airless) tires, but they should be interchangeable with the fielded pneumatic tires. Provisions shall be made for securing the wheels to the heater when not installed for use. The heater shall have continuous handrails or minimum handles to assist in manual localized movement and positioning. The heater shall be provided with a parking stand to provide three-point support when the wheels are installed and with integral means to prevent movement of the heater when parked in level or specified inclined positions (see 3.7.6). Weight distribution of heater components shall provide a safe and stable configuration in all specified operating positions.

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3.8.4, lines 3 and 4, delete "within 15-seconds" and substitute: "within 18-seconds".

3.8.5, delete entire paragraph and substitute:

"3.8.5 High temperature limit. A high temperature limit control shall shut down the burner when the temperature at the heater discharge outlet exceeds the pre-set limit and illuminate the panel indicator light. After a high temperature limit shutdown, the flame failure control (see 3.8.4) shall lock out the burner and require reset from the control panel. The high temperature limit control shall reset automatically prior to the outlet

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temperature falling to 125 °F. The pre-set limit on the high temperature limit control shall be 200 +/- 10 °F."

3.8.7, Add the following to the beginning of the paragraph: "The heater shall incorporate carbon monoxide (CO) sensors (primary and backup) which, when activated, will cause an audible and a visual alarm to be activated and will shut down the burner. The visual alarm shall remain on after system shutdown until it is reset. The primary sensor shall be in the air discharge (supply) outlet of the heater, and the backup sensor shall be located on the remote control box."

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3.9, add the following to the end of the paragraph: "The control circuit shall include a method of monitoring and controlling flame quality, controlling valves, solenoids, and relays. Displays and indicators must be simple, straightforward, and easily read or recognized."

3.9.5, delete in its entirety and substitute:

"3.9.5 Remote control box. A hermetically sealed remote control box shall be provided. The control box shall contain a thermostat with a manual override and a minimum range of from +35 to +85 F. The thermostat will provide a maximum differential of +/-5 F [T], +/-3 F [O] from the setting which controls the heated shelter's temperature. The remote control box will permit a designated operator to start or stop the unit or change the thermostat setting by following the attached instruction plates. The box shall be assembled with a cable of 25 feet minimum length and an environmental resistant connector compatible with a mating receptacle on the heater. The box shall also contain a CO sensor (see 3.8.7)."

3.9.6, delete subparagraphs a. thru g. and substitute:

"a. The life accuracy of the meter shall be ±0.1 percent over all operating and storage conditions. The accuracy may be degraded over the frequency range from 50 to 70 Hz, at the same percentage as the percentage change in the input frequency from 60 Hz.

b. The total time range readout of the meter shall be 9,999 hours."

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3.10.1, at the beginning of the paragraph add: "The heaters will have detachable, flexible, insulated ducts for supply- and return-air openings (separate duct storage is

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acceptable). The heater ducts shall be sized for compatibility with standard shelter openings."

Line 2, delete "with two 15-foot long ducts." and substitute "with two 12-inch nominal diameter, 15-foot long ducts."

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3.10.2, add at the beginning of the paragraph: "The heated air supply and return openings shall be 11.75 ±0.125 inches in diameter."

3.11.2, line 2, delete "color of the heater shall be Camouflage Green" and substitute "color of the heater shall be camouflage pattern in accordance with 3.11.2.4. Top coat colors for the camouflage painting shall be specified on the camouflage patterns. First Articles need not be camouflage painted."

3.11.2.2, last line, delete "treated, but shall not be painted." and substitute: "treated, but does not need to be painted."

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3.13, at the beginning of the paragraph add: "The ASH will be operable and maintainable in daylight and darkness by representative soldiers dressed appropriately for the anticipated environments for use (arctic, NBC, etc.). Risk to personnel will be minimized by design and performance characteristics. Safety deficiencies will be identified and eliminated or mitigated to acceptable levels. This will be verified by analysis, inspection, and test. Heater design will be IAW applicable Army safety standards. The heaters will not present uncontrolled health hazards throughout the life cycle of the system. All components will be located to permit easy access for operating and maintaining the heater. The ASH shall be designed and/or configured to provide an efficient and effective user interface for all critical install, operate, and maintain (IOM) tasks or procedures."

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3.14.1, add at the end of the paragraph: "In addition, the ASH must have a probability of completion of 20 consecutive days operating 21.5 hrs per day with no more than one essential function failure/system abort of at least R(EFF/SA)=0.94. The Median Time To Repair (MedTTR) for all Essential Unscheduled Maintenance Demands (EUMD) must not exceed 1 hour. The Maximum Time To Repair for 90% of all EUMD (MAXTTR 90) must not exceed 1.5 hours."

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3.14.3, delete first sentence and substitute: "The total maintenance ratio (MR) must not exceed 0.013 maintenance man-hours / operating hour."

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3.15.2, last sentence, delete "located on the remote control box," and substitute "located on the control panel,".

3.15.3, delete in its entirety and substitute:

"3.15.3 "Preventive maintenance checks and services (PMCS). Under normal 24-hour operating conditions (three 8-hour shifts), the heater will provide 21.5 hours of on-demand heat. The additional 2.5 hours will consist of 1.5 hours for set up (to include starting, shut down, and relocation), and 1 hour for PMCS. The heater will be used for varying time periods with refueling and scheduled services performed as required. Routine servicing will be accomplished at the operator/crew level and will consist of, but not be limited to, cleaning and replacing filter elements, refueling, and conducting PMCS."

3.15.5, at the end of the paragraph add: "The heater shall be repairable by module replacement of major component systems."

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4.5, b, line 2, delete "between -40 and +65 °F " and substitute "between -60 [T], -65 [O] to +65 °F ".

Table II: Under column "Requirements Paragraph", for "Altitude" Test: delete: "3.3..1, 3.4.1" and substitute: "3.3.1, 3.4.1".

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Table II, line 8 (as shown), delete row for "EMI" test in its entirety and substitute:

X								EMI and HEMP	4.6.8	3.4.7, 3.5, 3.5.1, and 3.9

4.6.2, line 2, delete " between -40 and +65 °F " and substitute " between -60 [T], -65 [O] to +65 °F ".

4.6.3, line 3, delete “at sea level atmospheric pressure.” and substitute: “at 4,000 feet atmospheric pressure.”.

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4.6.4.1, Table III: Under column "Temperature °F" for "1400" Time: delete "114" and substitute: "145".

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4.6.4.2, lines 2 and 3, delete "test at -60 °F ambient temperature . . . tests at -40 °F.", and substitute: "test at -65 °F ambient temperature and manipulation and operation tests at -60 [T], -65 [O] °F."

4.6.4.2.5, Steps 5 and 6, delete "-60 °F" and substitute: "-60 [T], -65 [O] °F".

Steps 8, 9, 10, 11, and 12: delete "-40 °F" and substitute: "-40 [T], -60 [O] °F unassisted start, -60 [T], -65 [O] °F assisted start."

Step 11, also delete “JP-8” and substitute “DF-A”.

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4.6.4.7, first sentence: delete "temperature between -40 and +65 °F," and substitute: "temperature between +40 and +65 °F,".

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4.6.7, add after the first sentence: "The rate of fuel consumption of the heater shall be measured and recorded in order to determine the total amount of fuel consumed by the heater during the test."

Last line: delete "Failure to start and operate continuously for the specified period shall" and substitute: "Failure to start and operate continuously for the specified period, or nonconformance to 3.6, shall".

4.6.8 and all subparagraphs under 4.6.8, delete in their entirety and substitute:

"4.6.8 Electromagnetic interference and high-altitude electromagnetic pulse. The heater shall be subjected to the radiated emissions, electric field, 10 kHz to 18 GHz test (RE102) and the conducted emissions, power leads, 10 kHz to 10 MHz test (CE102) in accordance with MIL-STD-461, and as specified herein.

4.6.8.1 Radiated emissions test conditions.

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4.6.8.1.1 Operating Modes. The controls of the heater shall be set for the following modes of operation at steady-state conditions:

a. "VENT" mode.
b. "AUTO HEAT" mode, highest temperature setting.
c. "AUTO HEAT" mode, lowest temperature setting.
d. "MANUAL HEAT" mode.

4.6.8.1.2 Item orientation. The radiated emissions test shall be performed in its entirety as specified herein with each of the four sides of the heater facing the antenna.

4.6.8.2 Conducted emissions test conditions.

4.6.8.2.1 Operating Modes. The controls of the heater shall be set for the following modes of operation at steady-state conditions:

a. "VENT" mode.
b. "AUTO HEAT" mode, highest temperature setting.
c. "AUTO HEAT" mode, lowest temperature setting.
d. "MANUAL HEAT" mode.

4.6.8.2.2 Power lead selection. The conducted emissions test shall be performed in its entirety as specified herein on the power cable as a whole, and on each of the conductors in the power cable (hot, neutral, and ground). It need not be conducted with the lead assembly installed.

4.6.8.3 Failure criteria. Nonconformance to 3.5.1 shall constitute failure of these tests.

4.6.8.4 High-altitude electromagnetic pulse. The heater shall be subjected to the HEMP test using the apparatus, instrumentation, and procedures of MIL-STD-2169B (see 6.9). Nonconformance to 3.4.7, 3.5, 3.9, or MIL-STD-2169B shall constitute failure of this test.

4.6.10.1, step 2, delete "-40 °F (+0, -10 °F)." and substitute: -60 [T], -65 [O] °F (+0, -10 °F)."

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4.6.11.1, line 2, second sentence, delete "temperature between -40 and +65 °F.", and substitute: "temperature between +40 and +65 °F."

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4.6.15.1, line 2, delete "-40 and +65 °F", and substitute: “between -60 [T], -65 [O] and +65 °F."

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4.6.15.3, under "Maintainability" delete: "Accept if MR is 0.042 or less. Reject if MR exceeds 0.042." and substitute: "Accept if MR is 0.013 or less. Reject if MR exceeds 0.013."

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Add new paragraphs:

"6.8 Lead assembly. The purpose of this lead assembly is to allow the heater to be connected directly to the terminal lugs on military generator sets. In other words, it serves as a "pigtail" connector between the heater and the generator set, and allows the user to use the power connection method that is most convenient."

"6.9 HEMP Requirements. HEMP requirements and test procedures are classified "Secret." Contractors and their personnel must have the proper security clearances to participate in any procurement using this document. This test will be performed by the Government."

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Figure 1, delete and substitute new figures 1-a and 1-b.

Custodian:
Army - CR4

Preparing activity:
Army - CR4

Review activity:
DLA - IS

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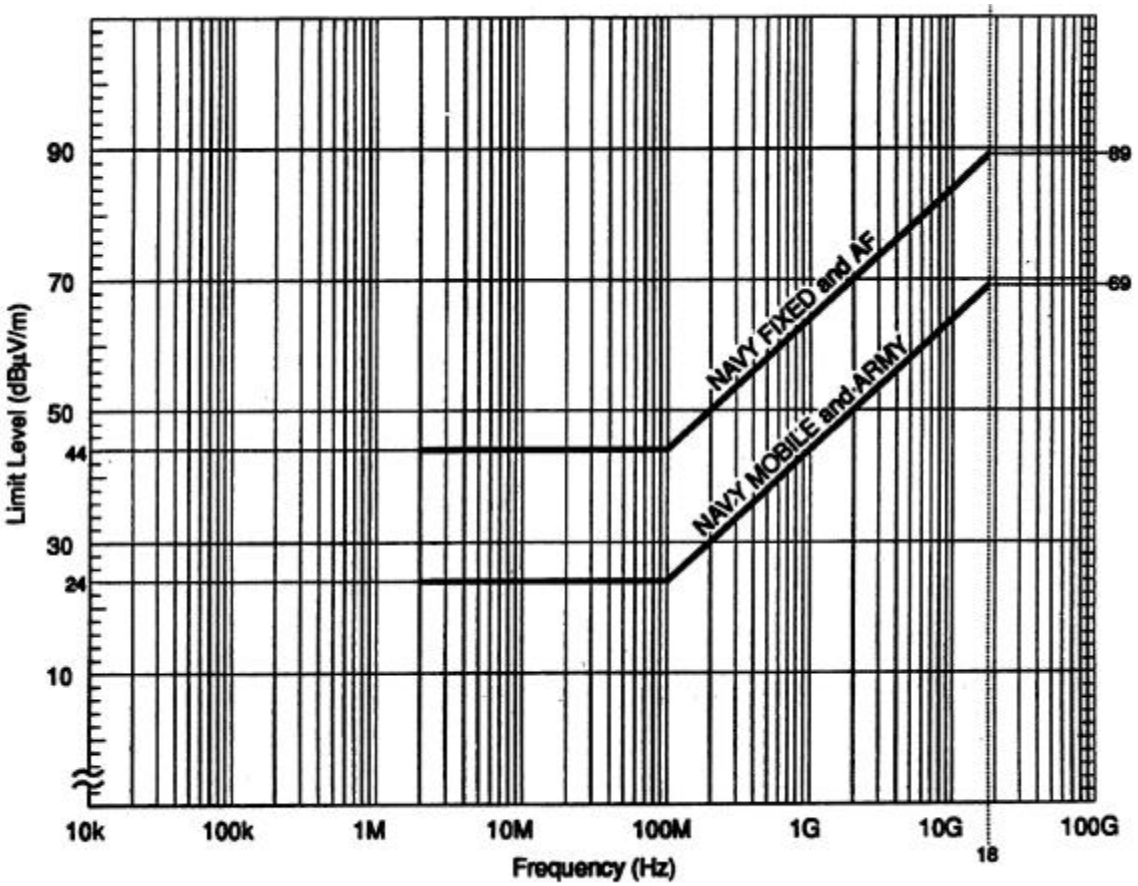


FIGURE 1-a RE102 LIMIT.

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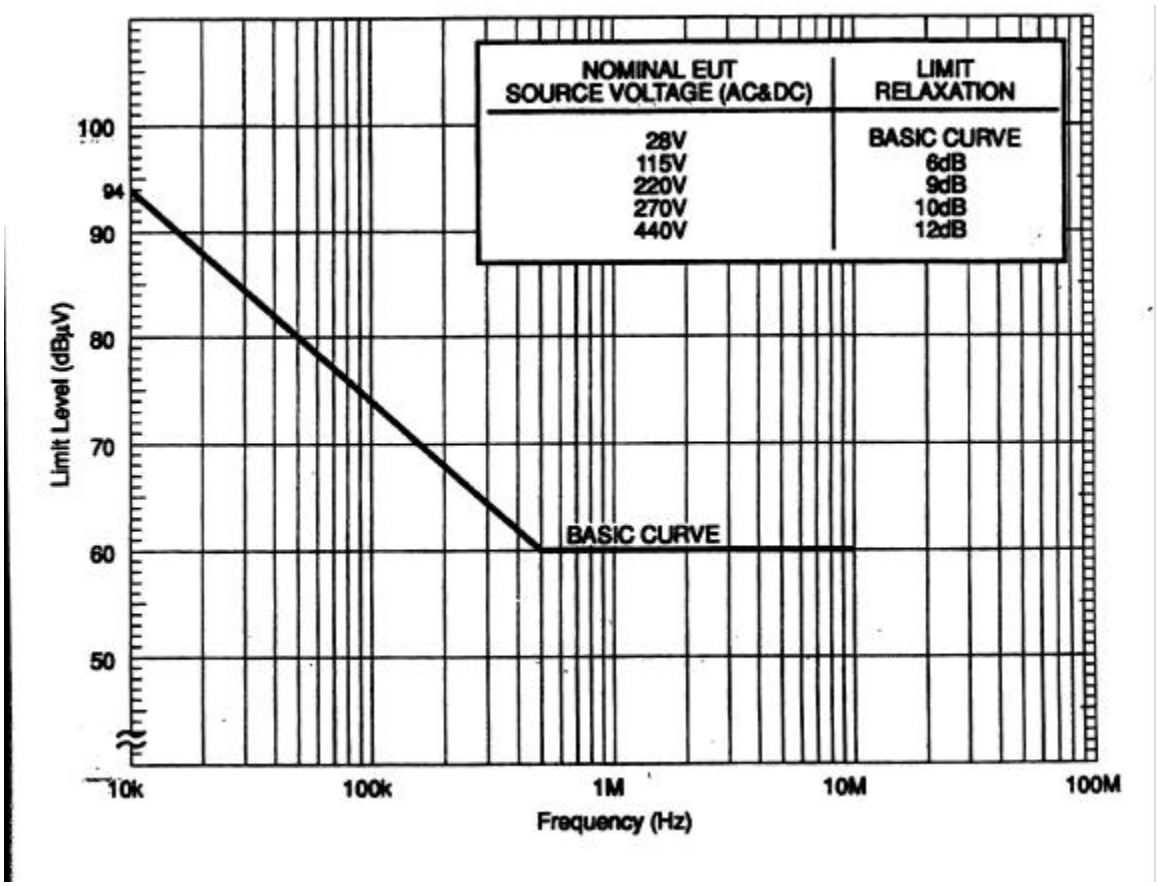


FIGURE 1-b. CE102 LIMIT

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